Lesson 5: Design a Carrier

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

One quick check before the lesson (2 minutes). Find the class tray with last week's **dried layered paper strips** (each has a pair's initials in pencil on the end) and put it where you can reach it. If the strips cannot be found, skip to the note at the bottom of this plan. Everything else is handed out by student helpers during the lesson. **No screens** are used.

This week each pair tests a layered strip against a single strip, then draws a carrier for a toy. Each student glues a record sheet into their **STEAM journal**. Journals are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:

To design a paper carrier for a toy by choosing techniques that make paper stronger.

Success Criteria:

I can:

- test a layered strip and a single strip to see which holds more.
- draw a carrier for my toy with a base, sides and a handle.
- label the techniques I will use.

Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
Layered strips from Lesson 4	1 per pair (on the class tray)	The strip to test
A4 scrap paper	3 sheets per pair, plus 1 per student	1 sheet to tear a single strip, 1 sheet for the design, 1 spare, and a record sheet each
Books	2 per pair	Making a gap about as wide as a hand for the strip to sit across
Small blocks or counters	About 30 per pair, in a cup	Adding weight, one at a time (Unifix cubes, counters, large paper clips or anything small and the same)
A small toy	1 per pair	The toy the carrier is for. Any small class toy or soft toy, or a pencil case if there are no toys.
Pencils and rulers	1 each per pair	Drawing and measuring the toy
Glue sticks	1 per pair	Gluing the record sheet into journals

STEAM journals	1 per student	Shelves by the door

Students tear the single strip themselves from scrap paper. Show them the length of the layered strip so the single strip is about the same length and width.

How to test the strips and design the carrier

1. Make the gap

Both ends rest on a book.

2. Add blocks

Write down the number.

3. Draw the carrier

Side view. Label two techniques.

- **Predict first:** each pair says which strip will hold more blocks, the single strip or the layered strip.
- **Make the gap:** stand two books about one hand apart (10 cm). Lay a strip across so each end rests on a book.
- **Add blocks:** place blocks one at a time in the middle of the strip, counting aloud. Stop when the strip sags and touches the table, or falls through. Write the number down.
- **Test the other strip:** same gap, same way. Compare the two numbers.
- **Choose the toy:** measure it with a ruler or hand spans. The carrier must be big enough to hold it.
- **Draw the carrier:** from the side, with a base, sides and a handle. Pencil first.
- **Label the techniques:** use at least two from the unit: folding, twisting, weaving or pasting. Draw an arrow to the part and write the technique, for example "woven sides" or "twisted handle".

Key words

design, carrier, technique, layer, stronger, handle, base

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Ask: *"Last week, what did we make by mixing flour and water?"* (Paste, a new material.) *"What did we do with the paste?"* (Stuck strips together in layers.) Read the Learning Intention and Success Criteria aloud.

3 to 8 min	Hook	Hold up a single strip of paper and a dried layered strip. Ask: *"Which will hold more blocks?"* Thumbs up for the single strip, thumbs down for the layered strip. Stand two books a hand apart and lay the single strip across. Add blocks one at a time while the class counts aloud, until it sags to the table. Say the number. Say: *"Now the layered strip. Is the paste doing something?"* Test it the same way and compare the numbers. Ask: *"What changed?"* (The layers are stuck together so they work as one stronger strip.)
8 to 18 min	Test in pairs	Helpers hand out a layered strip (the pair's own, with their initials), a sheet of scrap paper, two books and a cup of blocks per pair. Each pair tears a single strip the same length and width as the layered one, predicts, tests both and writes both counts on a record sheet. Roam with the questions below.
18 to 23 min	Show how to design	Write **design**, **carrier** and **technique** on the board and say what each means (a design is a plan for something we will make; a carrier holds and carries things; a technique is a way of doing something, such as folding, twisting, weaving or pasting). Hold up a toy and say: *"Our job is to make a carrier that will hold this toy. It must be strong."* On the board, draw a carrier from the side: a base, two sides and a handle. Add labels with arrows, for example "woven sides, because weaving was strong" and "pasted base". Say aloud why each technique was chosen.
23 to 38 min	Design in pairs	Pairs choose a toy and measure it. They draw their carrier on a sheet, with a base, sides and a handle, and label at least two techniques. Pencil first. Ask each pair to write one sentence about why they chose each technique. Roam with the questions below.
38 to 45 min	Share	3 or 4 pairs share their design using the sentence stem on the board. Come back to the Learning Intention and Success Criteria. Finish with the key idea (below). Helpers hand out journals. Students glue in their test record sheet and design, and write the date and the title **Design a Carrier**. Journals onto the shelves by the door.

Write this on the board

	Single strip	Layered strip
How many blocks did it hold?	_____	_____
Which was stronger?		

Carrier labels: **base**, **sides**, **handle**, and an arrow to each technique (fold, twist, weave, paste).

Sentence stem: **Our carrier will be strong because we used _____ for the _____.**

Questions to ask while roaming

- Which strip do you think will hold more? Why?
- How many blocks did the single strip hold? What about the layered strip?
- What is the paste doing to the paper?
- Where will your carrier carry the weight? Which technique will help there?
- How will you make your handle strong?

Support and extension

Who	What to do
Support	Draw a base, sides and handle outline for the pair, so they only add the labels. Students can record with a drawing and a number instead of a sentence.
Extension	Fast finishers (often Year 2) add measurements to their plan (for example "the base is 3 hand spans long") and draw a second handle or side idea. They write which idea is stronger and why.

Key idea to finish on: Layered paper held more blocks than a single strip, because the paste joined the layers so they work together. A **design** is a plan for something we will make. Next week we will make our carriers from our plans, so a good label now makes the making easier.

Notes for the teacher

- **Strips still damp?** A damp strip is weaker. Let that pair test it last and write "damp" next to the number. That is a useful result, not a mistake.
- **Strips missing?** Each pair can instead tear and paste a quick layered strip with a glue stick (3 strips, glue between). It will be weaker than a pasted one, but the comparison with a single strip still works.
- **Fair test:** both strips should be the same paper, the same length and rest on the same gap. Check the gap with a ruler.
- **Falling blocks:** blocks can spill when a strip folds. Test over a table, and keep the cups of blocks away from the edge.
- **Counts vary:** results will differ between pairs, which is fine. Look for the pattern across the room: most layered strips held more.
- **Please keep every pair's design.** Collect them in a pile (or leave them in journals) because next lesson the class makes their carriers from them.
- Please leave a short note about how far the class got and any students who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2D01 to VC2TDE2D03:** generate and communicate design ideas, and plan the steps to make a designed solution.
- **Maths:** counting blocks, measuring with a ruler or hand spans.